

What Do Wrist-Worn Stress Detectors Detect? An Untrained Arousal Index Nearly Matches Them Across Five Empatica E4 Datasets

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Abstract—Wrist-worn stress detectors are trained to separate stressor periods from rest and are reported as detecting stress. We ask what they detect. On five public laboratory Empatica E4 datasets (132 subjects) with audited labels and subject-grouped leave-one-dataset-out evaluation, a heart-rate, heart-rate-variability and electrodermal model trained on four datasets loses 0.015–0.05 balanced accuracy on the fifth, none significant. An untrained index of heart rate and tonic electrodermal activity comes within 0.075 of the model on every unseen dataset and is statistically equivalent to it on two. At matched arousal, the model still separates stress from rest on WESAD (AUROC 0.95) but only modestly on the other four datasets (0.57–0.70), and no better from a lower-demand social task, hyperventilation or manual tasks than from rest; these nulls are robust across normalisations and quality screening but can exclude only deviations above 0.12, and the anger and hyperventilation cells are small and exploratory. Exercise is called stress in 69% of windows until it is added to training. Protocol position alone predicts stress better than physiology on the two hardest datasets, whose baselines fall in the first minutes of recording. Domain adaptation, few-shot personalisation and per-user thresholds add nothing significant. On current public E4 data, wrist stress detectors behave largely as arousal detectors. We propose an index-only baseline, a specificity panel and a matched-arousal test as reporting checks, and release audited labels, frozen subject splits and code.

Index Terms—Wearable stress detection, Empatica E4, autonomic arousal, cross-dataset generalization, specificity, electrodermal activity, heart rate variability.

I. INTRODUCTION

STRESS detection from wrist wearables is usually posed as a classification problem: windows recorded during a laboratory stressor are labelled stress, windows recorded during rest or a control task are labelled non-stress, and a model is trained to separate them. Reported accuracies on the standard datasets are high, and the models are described as detecting stress. Two observations motivate this paper. First, the cross-dataset literature disagrees about whether such models transfer: Prajod et al. [1] and Calza-Metre and Borzi [2] report large drops when the stressor changes, Xiao et al. [3] report balanced accuracy falling from 0.88 on WESAD to 0.53–0.69 on three out-of-distribution datasets, whereas Mishra et al. [4] found a cost of about 0.01 between two Empatica E4 studies with the same stressor, and Kwon et al. [5] found near-perfect transfer to WESAD but near-chance results on Stress-Predict. Second, the wrist device measures pulse rate,

a weak vagal proxy, wrist electrodermal activity (EDA), skin temperature and movement. None of these is stress; all of them respond to autonomic arousal from any cause. Kwon et al. [5] matched heart rate within subject and found that WESAD stress remains separable from baseline through EDA and temperature, which rules out a heart-rate shortcut there. Whether a detector separates stress from other arousing non-stress states when heart rate and EDA are matched together has not, to our knowledge, been tested.

We address both observations with the same five public E4 datasets. After auditing every dataset’s labels against its descriptor paper, validating wrist cardiac features against chest electrocardiography (ECG), and fixing an evaluation protocol with subject-grouped splits and corrected tests, we show that the transfer cost is small, with or without per-subject normalisation and label fixes, and that the remaining disagreement in the literature is largely about absolute difficulty rather than transfer. We then ask what transfers. An untrained arousal index is scored under the same leave-one-dataset-out protocol as the model; a specificity panel scores the fixed detector on states absent from training; and a matched-arousal test asks whether stress stays separable from those states, and from rest, at equal arousal. A time-only classifier, a sensor-settling analysis and a test-side error budget then locate the remaining within-dataset ceiling.

The contributions are two. (1) A five-dataset leave-one-dataset-out benchmark with audited protocol-stage labels, showing a small, non-significant transfer cost (0.015–0.05 balanced accuracy, upper bounds 0.08–0.13), diagnosing one apparent domain gap as a missing channel, and quantifying time in session and early-recording drift as confounds. (2) Evidence that the trained detector behaves as an arousal detector: an untrained heart-rate and EDA index comes within 0.075 BA of it on every unseen dataset, outside WESAD, stress is only modestly separable from rest at matched arousal and no better from lower-demand tasks, and exercise is called stress until it is in training; from these we derive three reporting checks, an index-only baseline, a specificity panel and a matched-arousal test with a reference pair. An error budget and leakage-controlled negative results are in the supplement. Labels, frozen splits and scripts are released. Section II reviews related work, Section III describes the datasets and the label audit, Section IV the features, normalisation and evaluation rules, Section V the results, and Section VI discusses implications, reporting norms and limitations.

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II. RELATED WORK

A. Cross-dataset transfer

Kwon et al. [5] ran pooled leave-one-dataset-out (LODO) evaluation on three E4 corpora with logistic regression and gradient boosting and no per-subject normalisation, obtaining an area under the receiver operating characteristic curve (AUROC) of about 0.90 with WESAD as target but 0.52–0.57 on Stress-Predict and the Nurse dataset; their within-dataset AUROC on Stress-Predict was already 0.63, so the transfer cost there is 0.06–0.08. Calza-Metre and Borzì [2] report pairwise zero-shot transfer across four E4 corpora, with WESAD macro-F1 falling from 0.87 within to 0.63 from Campanella. Xiao et al. [3] train on one or two of seven wrist datasets and observe balanced accuracy 0.53–0.69 out of distribution. Mishra et al. [4], with per-participant normalisation and matched arithmetic stressors, found a median AUROC cost of about 0.01. Schreiber and Maleshkova [6] pool three datasets under leave-one-subject-out (LOSO) evaluation and name LODO as future work. Prajod et al. [1] attribute transfer failure to stressor type, on chest ECG; their failing pair also differed in stress intensity, which they note “might have contributed to the poor performance”. Our reading of this literature is that the reported failures come from low within-dataset ceilings, from stressor mismatch, or from the absence of per-subject normalisation, and the present study tests each.

B. Label quality and confounds

Several authors name label noise as a cause of poor transfer [5], [7] but none audits and fixes labels and measures the effect. Mishra et al. [4] exclude post-stressor rest because it carries residual arousal; Aydoğan and Villagra Povina [8] found independently that the PhysioNet stress sessions mix rest and stressor blocks. Richer et al. [9] note that comparing a stressor to a pre-stress baseline confounds the comparison with sequence effects. Tognotti et al. [10] show that test-inclusive baseline normalisation inflates balanced accuracy by 3–13 points, and Fecke and Rehof [11] normalise from a calibration baseline on chest ECG. Aydoğan and Villagra Povina [8] report that an XGBoost model labels 82.6% of PhysioNet exercise windows as stress, and 6–7% once exercise is a training class. Saeb et al. [12] and Bahameish et al. [13] document the inflation from record-wise cross-validation. Schmidt et al. [14] recommend leave-one-subject-out validation and treat physical activity as a confound for which it is “sufficient to estimate the intensity level of an activity”. Vos et al. [15], found that apart from two studies, including Mishra et al. [4], none of the reviewed work tested its models on a totally unseen dataset, and cite the power analysis of Iqbal et al. [16], which concluded that “at least 34 test subjects would be required to achieve 80% statistical power”.

C. Personalisation and the arousal question

Adding a few labelled windows of a new user to a generic model is well established [17], [18], [19], [20]. Where calibration windows are taken chronologically, gains are small:

Tervonen et al. [19] found retraining XGBoost gained on average 1.7 balanced-accuracy points across datasets and tasks with the first 1–3 samples of each stimulus, and the abstract of Akkaya [20] reports that the few-shot benefit did not survive a prospective chronological protocol. On wrist cardiac validity, Milstein and Gordon [21] found that the root mean square of successive differences (RMSSD) from the E4 correlated with ECG at $r = 0.42$ during a dyadic conversation against 0.74–0.81 at rest, and Watanabe et al. [22] found E4 beat detection during the WESAD stress task, a Trier Social Stress Test (TSST), more sensitive to filter settings than at rest. Sosa et al. [23], reviewing multimodal arousal classifiers, state that “physiological signals primarily index arousal and regulatory processes rather than affective valence” and that sympathetic activation “can therefore arise not only during stress, but also during non-stressful states”; most reviewed studies did not test classifiers on such states. Kwon et al. [5] balanced heart rate within 5-bpm bins per subject: on WESAD the all-feature AUROC moved from 0.955 to 0.938 and EDA plus temperature alone kept 0.935, while on Stress-Predict every matched feature set stayed weak (EDA plus temperature 0.607). Their contrast is stress against baseline; the matched-arousal test below instead matches heart rate and EDA jointly and contrasts stress with other arousing states.

The gap is therefore twofold. No study has scored one fixed, subject-independent wrist detector across a panel of non-stress arousal states, or asked whether stress remains separable from those states at equal arousal; and no study has decomposed the within-dataset ceiling of wrist stress detection into recovery, sensor, onset, responder and stressor terms. The present paper addresses both on five public datasets with audited labels.

III. DATASETS AND LABEL AUDIT

Five public datasets recorded with the Empatica E4 (blood volume pulse (BVP) 64 Hz, EDA 4 Hz, skin temperature 4 Hz, accelerometer 32 Hz) contain both stress and non-stress periods: WESAD [24] (15 subjects; TSST speech and arithmetic before a panel, standing), the PhysioNet stress and exercise dataset of Hongn et al. [25] (35; Stroop, the Trier Mental Challenge Test (TMCT) of mental arithmetic under time pressure and serial subtraction, seated, plus aerobic and anaerobic sessions; its opinion tasks yield no complete window), Stress-Predict [16] (34 after exclusion; Stroop and a TSST-style interview, plus a hyperventilation block that its descriptor lists among the stress-inducing tasks), UBFC-Phys [26] (19; speech and arithmetic in a harder “test” version, a job interview and a countdown in steps of 17, or an easier “control” version, a holiday speech and a countdown in steps of 10, both spoken aloud to the experimenter; the authors note that both versions may “imply a high stress level”) and Campanella 2024 [27] (29; backward subtraction, plus three Lego assembly tasks designed to create work-like “mental strain”, one with a countdown; the closing CV presentation lies in the untaged tail and is excluded). EPM-E4 [28] (33 subjects with clip timing) supplies film-induced anger, fear, happiness and sadness and is used only as a probe. Participants were young adults in WESAD (mean age 27.5, 12 of 15 male),

TABLE I
HARMONISED CLASSES (SUBJECTS / 60-S WINDOWS AT 30-S STEP)

Dataset	Original label → harmonised class	Subj./win.
WESAD	TSST → Stress	15 / 311
	Baseline; Meditation; Amusement	15 / 566; 356; 165
PhysioNet	Stroop, TMCT, subtraction → Stress	35 / 232
	Baseline (3 min before first tag); First and second rest	35 / 174; 1193
	Aerobic; Anaerobic (whole sessions)	30 / 2143; 31 / 1620
	Stroop; TSST interview → Stress	33 / 851
Stress-Predict	Baseline; Relax → Rest	34 / 616; 1079
	Hyperventilation (probe only)	31 / 53
	T2, T3 test group → Stress	11 / 110
	T1 rest → Baseline	19 / 95
UBFC-Phys	T2, T3 control group → Control task (probe)	8 / 80
	Backward subtraction → Stress	29 / 102
	Baseline	29 / 116
	Lego tasks → Manual task (probe)	29 / 667
Campanella	Anger; Fear; Happiness; Sadness (probes)	33 / 33; 394; 297; 131

PhysioNet (19–30 years) and UBFC-Phys (mean 21.8 for the full cohort of 56), older in Stress-Predict (mean 32, 10 of 35 male) and Campanella (20–60 years, 21 of 29 male); EPM-E4 reports neither age nor sex. Table I gives the harmonised classes.

Every labelling decision was checked against the dataset’s descriptor paper. Seven corrections followed: in WESAD, labels had to be taken from the synchronised file rather than mapped by length onto the raw E4 export, whose recording starts 16–26 min earlier; in PhysioNet, only the tagged stressor tasks count as stress, not the whole stress session, one subject with a documented bad fit (f13) is excluded, and the 3 min before the first tag serve as baseline; in Stress-Predict, subject S01 violated the protocol and is dropped, and task boundaries whose tags are more than 180 s from the logged time are dropped; in Campanella, which logs no per-subject task markers, the subtraction span ends at a per-subject end estimated from the recording length. In addition, the Stress-Predict descriptor tags its 2-min hyperventilation block as a “stress-inducing task”; it is a respiratory manoeuvre with no evaluative component, and we keep it as a separate probe class. The three participants recorded in two files are treated as one subject each, so that no person appears on both sides of a split.

A cautionary note on evaluation. An earlier version of this pipeline used window-level train/test splitting with 50%-overlapping windows and reported 94% accuracy on a six-class task (92% in our reproduction); the same model scores 49% (balanced accuracy 0.36) when 15% of subjects are held out (`leakage_check.csv`). Record-wise splitting [12] remains common in this field, and every number below uses subject-grouped splits.

IV. METHODS

A. Features

Windows are 60 s at a 30-s step, anchored inside one protocol stage. Cardiac features, the mean and standard deviation (SD) of heart rate (HR) and three heart-rate-variability (HRV) measures (RMSSD, the SD of normal-to-normal intervals (SDNN) and the proportion of successive differences above 50 ms (pNN50)), come from BVP beats detected with NeuroKit2 after band-pass filtering and artefact rejection; windows less than half covered by accepted beats carry missing cardiac values. Against WESAD chest ECG on the same 1,398 windows, wrist heart rate has a mean absolute error of 2.4 bpm ($r = 0.94$); RMSSD is weak ($r = 0.53$, bias +34 ms), and only 37% of TSST windows keep usable photoplethysmography (PPG) because speech moves the wrist (heart-rate bias −7 bpm there). EDA features are computed on the raw 4-Hz signal of each window without artefact rejection: mean, SD, range and slope; a tonic component from a zero-phase second-order Butterworth low-pass at 0.05 Hz, with the phasic mean as the mean absolute residual; and skin conductance response (SCR) count and mean amplitude from peaks whose prominence exceeds half the SD of the window’s first difference, at least 1 s apart. A quality screen with the rules of Kleckner et al. [29] is reported as a sensitivity analysis. Temperature features are included only in the “physiology” set; the main set is HR/HRV/EDA, for reasons given in Sections V-B and V-E. Accelerometer features are used only in the modality comparison.

B. Normalisation

Features are z-scored per subject using all windows of that subject’s stress-protocol recording, without labels. This is transductive; two non-transductive variants are reported as sensitivity analyses: raw features, and causal scaling from strictly earlier windows only (the first four windows share the statistics of that opening period).

C. Task, models and evaluation

The main task is stress versus all non-stress states (baseline, rest, meditation, amusement). Baseline-versus-stress alone is confounded with time in session (Section V-E). The classifier is XGBoost with fixed hyperparameters; a nested-tuning comparison of XGBoost, random forest and a multilayer perceptron found one significant difference in 72 comparisons after Holm correction, on the nine-class affective task only and none on the stress tasks (`significance_models.csv`), so tuning is not repeated. Evaluation uses 20 subject-grouped splits with 20% of each dataset’s subjects held out, LOLO where per-subject scores are needed, and LODO in which the target’s held-out subjects are scored by a model trained (a) on the target’s other subjects, (b) on the other four datasets only, or (c) on both. Paired comparisons use Nadeau–Bengio corrected t -tests [30] with Holm correction; per-subject rates carry subject-bootstrap 95% confidence intervals (CI). The threshold is 0.5 throughout; balanced accuracy (BA) and AUROC are reported.

D. Four evaluation rules

Specificity panel. A fixed detector is scored on non-stress states absent from its task definition, reporting the false-stress rate ($p > 0.5$) under three training conditions: the state’s dataset absent from training; all datasets in training but the state itself absent; the state added to the training negatives. *Matched-arousal test.* A label-free arousal index (mean of per-subject z-scored heart rate and tonic EDA, or tonic EDA alone where heart rate is missing) is computed per window; stress windows are pooled with one non-stress arousal state, binned into arousal quintiles, and the AUROC of the stress score is computed within bins and averaged by bin size. An AUROC at the level of the reference pair means the model separates that state no better than it separates rest. Each probe pair is compared with the stress-versus-baseline/rest pair of its own dataset under the same model (Δ ref, joint subject bootstrap; for the EPM-E4 clips, whose stress windows come from all five datasets, the pooled pair with the same stress windows), tested against 0.5 by a within-bin permutation test with Holm correction, and reported with its minimum detectable deviation (80% power) and a ± 0.10 equivalence test. *Index-only baseline.* The same index, thresholded on the training datasets, is scored under LODO and compared with the model by paired Nadeau–Bengio tests. *Position-matched negatives.* Non-stress test windows are restricted to the 5th–95th percentile range of minutes-since-start spanned by that dataset’s stress windows. *Test-side error budget.* Filters applied to out-of-sample predictions without retraining, with gains Shapley-averaged over all filter orders.

V. RESULTS

A. Subject-independent performance

On the pooled baseline-versus-stress task of WESAD and PhysioNet (50 subjects) with raw (unnormalised) features and the fixed-hyperparameter model, 20 subject-grouped splits give BA 0.83 (Nadeau–Bengio 95% CI 0.79–0.88; `modality_nb_ci.csv`) with all modalities, 0.81 (0.76–0.86) with physiology only and 0.75 (0.68–0.82) with the accelerometer only; LOSO agrees (0.85, 0.81, 0.75; `loso_only_summary.csv`). No modality difference survives Holm correction ($p_{\text{holm}} = 1.0$, `significance_modalities.csv`), and a complete 108-configuration nested tuning run changes none of these conclusions. The model is therefore not detecting movement alone, and physiology carries most of the signal. On the main task and all five datasets, within-dataset BA with HR/HRV/EDA features is 0.916 (WESAD), 0.882 (Campanella), 0.877 (UBFC-Phys), 0.776 (PhysioNet) and 0.697 (Stress-Predict) (Table II). The two lowest ceilings belong to the two datasets whose stressors provoke the smallest electrodermal response: mean EDA rises by 0.4 z in PhysioNet and 0.7 z in Stress-Predict against 1.7–2.2 z in the other three (`stress_effect_size_all_datasets.csv`).

B. Cross-dataset transfer

Table II and Fig. 1 give the LODO result. With HR/HRV/EDA features, a model trained on the other four

TABLE II
LEAVE-ONE-DATASET-OUT BALANCED ACCURACY ON HELD-OUT SUBJECTS, STRESS VS ALL NON-STRESS (MEAN OF 20 SPLITS)

Target	HR/HRV/EDA model			Index	LR	+ temperature		
	Within	Other 4	All 5			Within	Other 4	All 5
WESAD	0.916	0.884	0.921	0.836	0.824	0.900	0.840	0.876
PhysioNet	0.776	0.746	0.783	0.743	0.723	0.760	0.735	0.770
Stress-Predict	0.697	0.646	0.689	0.657	0.657	0.712	0.645	0.699
UBFC-Phys	0.877	0.837	0.839	0.762	0.756	0.877	0.589	0.867
Campanella	0.882	0.867	0.895	0.877	0.877	0.897	0.827	0.895

Index: the untrained arousal index (mean of per-subject z-scored heart rate and tonic EDA), thresholded at the value that maximises BA on the other four datasets. LR: logistic regression trained on the same two features on the other four datasets.

datasets loses 0.03 (WESAD), 0.03 (PhysioNet), 0.05 (Stress-Predict), 0.04 (UBFC-Phys) and 0.015 (Campanella) BA against the within-dataset model on the same held-out subjects. Training on all five datasets matches or exceeds within-dataset training on WESAD, PhysioNet and Campanella and is within 0.04 on the other two (Stress-Predict 0.689 against 0.697, UBFC-Phys 0.839 against 0.877). None of the five costs is significant after Holm correction (Nadeau–Bengio $p_{\text{holm}} \geq 0.19$; only Stress-Predict excludes zero before correction, 0.052 [0.003, 0.101]), but with 3–7 held-out subjects per split the upper 95% bounds reach 0.08–0.13 (`D_lodo_cost_corrected.csv`). We therefore claim no large cost, not equality.

An untrained arousal index comes close to the model. If the detector has learned arousal magnitude, a label-free arousal index should do nearly as well. The index (mean of per-subject z-scored heart rate and tonic EDA, EDA alone where heart rate is missing) needs no training; only its threshold is set, on the other four datasets. Table II gives its external BA: 0.836 on WESAD, 0.743 on PhysioNet, 0.657 on Stress-Predict, 0.762 on UBFC-Phys and 0.877 on Campanella, with AUROC 0.73–0.95. The trained model beats it by 0.048 on WESAD and 0.075 on UBFC-Phys, ties on PhysioNet (+0.003), and trails it on Stress-Predict and Campanella; no difference survives Holm correction (paired Nadeau–Bengio, $p_{\text{holm}} \geq 0.14$; `C_arousal_index_lodo.csv`). The differences are equivalent to zero within ± 0.05 only on PhysioNet and Stress-Predict (90% CIs inside the margin); on UBFC-Phys the model’s advantage of 0.075 has a 95% CI of [0.009, 0.140] before correction, and on WESAD the upper bound reaches 0.12. Learning the weights of the two features adds nothing: a logistic regression on heart rate and tonic EDA, trained on the other four datasets, is equivalent to the fixed index within ± 0.05 on all five targets (differences -0.02 to 0.00 ; `two_feature_lr/`), with weights of similar size for the two channels (EDA-to-heart-rate ratio 0.9–1.5), so the model’s remaining edge of up to 0.08 comes from its other features. An EDA-only index loses significantly to the model on PhysioNet (-0.15) and a heart-rate-only index on UBFC-Phys and Campanella, so neither channel alone suffices. A model trained on four datasets of stressor labels thus adds at most a few points of balanced accuracy on the fifth over an untrained two-channel arousal index, and on two datasets nothing measurable.

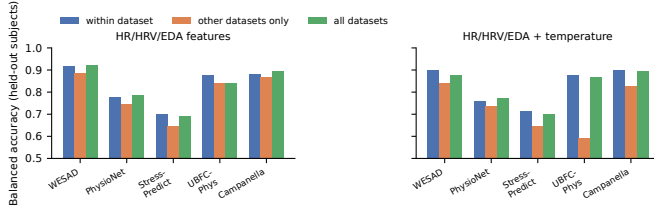


Fig. 1. Leave-one-dataset-out balanced accuracy on held-out subjects for the two feature sets. With temperature included, UBFC-Phys, which records no temperature, drops to 0.589 when trained on the other datasets although its AUROC stays at 0.931.

A missing channel posing as a domain gap. With temperature included, external UBFC-Phys falls to 0.589 BA while its AUROC stays 0.931 and an oracle threshold recovers 0.910 (`threshold_transfer_probe_summary.csv`). UBFC-Phys has no temperature in any window, so XGBoost’s missing-value branch pushes its mean stress score to 0.18 against 0.31–0.42 on the other targets. Ranking is intact; the threshold is not. Dropping temperature lifts external UBFC-Phys to 0.837. Elsewhere the remaining external loss is mostly ranking: an oracle threshold recovers only 0.02–0.05 with HR/HRV/EDA features, and a label-free cut matching the source stress rate hurts Campanella (0.867 to 0.732).

Normalisation. The small cost does not depend on transductive scaling. With raw features the external cost is -0.01 to 0.05 BA; causal scaling keeps it on WESAD (0.845 within, 0.841 external), PhysioNet (0.774, 0.744) and Stress-Predict (0.688, 0.665) but loses about 0.10 externally on UBFC-Phys and Campanella, whose recordings are short (7–11 windows per subject). Whole-session scaling mainly raises the absolute level on both sides. No external difference between variants survives Holm correction across 60 tests.

Label fixes. Every external score rose after the seven corrections (for example UBFC-Phys 0.790 to 0.837, whose own data did not change), but on the two datasets where a paired test is possible only 1 of 16 differences survives Holm correction (WESAD external AUROC, $+0.0075$, $p_{\text{holm}} = 0.021$). In the two-dataset setting of Kwon et al. [5] (WESAD and Stress-Predict), the fixes change transfer by at most 0.02; our pipeline nevertheless reaches external AUROC 0.67 on Stress-Predict against their 0.56, and per-subject scaling raises WESAD as target from 0.834 to 0.914 AUROC. The label audit is a correctness contribution; the small transfer cost comes from the features and the task definition, since it stays small with raw features and was 0.04–0.09 even with the pre-fix labels.

C. Specificity panel

The LODO task keeps only baseline, rest, meditation and amusement as negatives; other recorded states were never scored. Table III and Fig. 2(a) give their false-stress rates. Exercise is the one probed state that is unambiguously not stress. A model that never saw exercise calls 69–70% of PhysioNet exercise windows stress, matching the 71–83% of Aydoğan and Villagra Povina [8]; adding exercise to the training negatives collapses this to 8–10%, and the external PhysioNet drop is significant (BA 0.746 to 0.583, $p_{\text{holm}} =$

TABLE III
FALSE-STRESS RATE ON STATES ABSENT FROM THE TASK DEFINITION (HR/HRV/EDA; SUBJECT-BOOTSTRAP 95% CI; WINDOWS COUNTED IN THE DATASET-ABSENT RUN)

Probed state (subjects / windows)	Dataset absent	State absent	Added as negative
PhysioNet aerobic (30 / 2143)	0.69 [0.59, 0.80]	0.61 [0.52, 0.71]	0.10 [0.05, 0.15]
PhysioNet anaerobic (31 / 1620)	0.70 [0.61, 0.79]	0.58 [0.50, 0.66]	0.08 [0.05, 0.11]
UBFC-Phys control version (8 / 80)	0.55 [0.46, 0.66]	0.57 [0.48, 0.66]	0.56 [0.45, 0.67]
Stress-Predict hyperventilation (31 / 53)	0.42 [0.26, 0.59]	0.54 [0.36, 0.71]	0.55 [0.36, 0.72]
Campanella Lego tasks (29 / 667)	0.43 [0.38, 0.48]	0.42 [0.36, 0.48]	0.33 [0.27, 0.39]
EPM-E4 anger (33 / 33)	0.42 [0.27, 0.61]	0.44 [0.27, 0.61]	0.35 [0.18, 0.53]
EPM-E4 fear (33 / 394)	0.35 [0.25, 0.45]	0.33 [0.23, 0.43]	0.25 [0.17, 0.34]
EPM-E4 sadness (33 / 131)	0.37 [0.27, 0.49]	0.35 [0.24, 0.49]	0.30 [0.19, 0.42]
EPM-E4 happiness (33 / 297)	0.24 [0.14, 0.33]	0.25 [0.15, 0.38]	0.18 [0.09, 0.29]
Reference: in-task non-stress states	0.08–0.29	0.05–0.25	—
Reference: stress recall, by dataset	0.53–0.91	0.63–0.95	within -0.05

0.011). No other state behaves this way, but the other probes are not clean non-stress states. The UBFC-Phys control version is an easier social task that its authors expect to be stressful, the Campanella Lego tasks were built to induce work-like strain, the Stress-Predict descriptor lists hyperventilation among its stress-inducing tasks, and the EPM-E4 clips are passive viewing. Where self-reports exist they show uneven manipulations (`manipulation_check.csv`): the stress rating rose over each subject’s baseline by 1.5 points of 5 for WESAD’s TSST (at least one point in 80% of subjects) and by 2.4 of 10 for PhysioNet’s TMCT (89%), but by only 0.5 for its Stroop task (61%); EPM-E4 participants rated the anger, sadness and happiness clips 6.3–7.0 on the 9-point SAM arousal scale, close to WESAD’s rating of the TSST (6.8), and the fear clips 5.0. Stress-Predict, UBFC-Phys and Campanella provide no comparable ratings. The UBFC-Phys control version is called stress at 0.55–0.57 whether or not the model has seen it; hyperventilation rises from 0.42 to 0.54–0.55 once Stress-Predict is in training; Lego tasks fall only from 0.43 to 0.33; and the EPM-E4 clips fall by 0.05–0.10. Part of the exercise collapse is its size: subsampled to 80 windows, the size of the small probes, exercise is called stress in 19% of windows rather than 8%, still far below 69%; under raw and baseline-referenced scaling the drop is from 81–83% to 19–30%. Adding a probed state to the negatives changes stress recall by at most -0.05 (UBFC-Phys 0.74 to 0.70 with Lego added; PhysioNet 0.74 to 0.71 and Campanella 0.92 to 0.89 with the EPM-E4 clips added), and exercise negatives raise it (Stress-Predict 0.63 to 0.70, UBFC-Phys 0.74 to 0.79), so the failure is not a sensitivity trade-off. The hyperventilation (1.7 windows per subject), anger (one window per subject) and UBFC-Phys control (8 subjects) cells are small; Lego and the fear clips carry the statistics.

D. Matched-arousal test

Table IV and Fig. 2(b) give the result. The reference sets the scale, and it differs by dataset

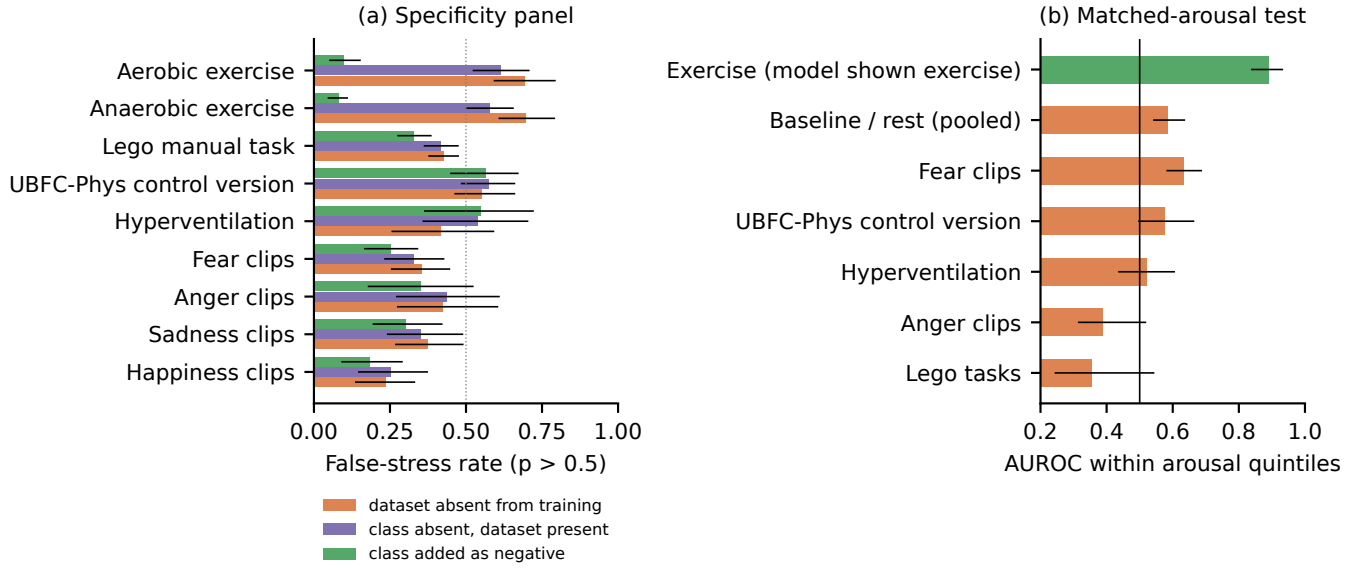


Fig. 2. (a) Specificity panel: false-stress rate of the HR/HRV/EDA detector on states absent from its task definition, under three training conditions, with subject-bootstrap 95% CIs. (b) Matched-arousal test: AUROC of the stress score within arousal quintiles for stress versus each state (external model; for exercise, the model that has seen exercise). Under session scaling only exercise and fear clips remain separable at equal arousal (exercise only under session scaling, fear clips not on windows with usable pulse; Section V-D). The baseline/rest bar pools all five datasets; per-dataset references (0.57–0.95) are in Table IV.

TABLE IV
MATCHED-AROUSAL TEST: AUROC OF THE STRESS SCORE INSIDE AROUSAL QUINTILES, AND ITS DIFFERENCE FROM THE SAME-DATASET STRESS-VERSUS-REST REFERENCE UNDER THE SAME MODEL

Stress versus	<i>n</i>	Within-bin [CI]	Ref.	Δ ref [CI]	p_{holm}
<i>External model (probed dataset absent from training)</i>					
Exercise	3585	0.759 [0.67, 0.83]	0.70	+0.06 [−0.02, +0.14]	0.25
Fear clips	394	0.633 [0.58, 0.69]	0.58*	+0.05 [0.00, +0.10]	0.16
UBFC-Phys control	80	0.577 [0.50, 0.67]	0.68	−0.11 [−0.36, 0.00]	0.16
Hyperventilation	53	0.523 [0.44, 0.61]	0.57	−0.05 [−0.13, +0.03]	0.25
Anger clips	33	0.390 [0.31, 0.52]	0.58*	−0.19 [−0.26, −0.06]	0.012
Lego tasks	667	0.355 [0.24, 0.54]	0.61	−0.25 [−0.43, −0.08]	0.035
<i>Model trained on all datasets with the probed state as a negative</i>					
Exercise	3585	0.892 [0.84, 0.93]	0.70	+0.19 [+0.13, +0.26]	<0.001
Fear clips	394	0.779 [0.74, 0.82]	0.72*	+0.05 [+0.01, +0.10]	0.044
UBFC-Phys control	80	0.568 [0.46, 0.67]	0.73	−0.17 [−0.41, −0.05]	0.044
Hyperventilation	53	0.560 [0.45, 0.67]	0.63	−0.07 [−0.18, +0.04]	0.45
Lego tasks	667	0.496 [0.44, 0.70]	0.63	−0.13 [−0.30, +0.16]	0.45
Anger clips	33	0.431 [0.33, 0.62]	0.72*	−0.29 [−0.39, −0.11]	<0.001

n: probe windows with a defined index (178 exercise windows lack both heart rate and EDA). Within-bin AUROC is weighted by bin size, with a subject-bootstrap CI (2000 draws). Ref.: stress versus baseline and rest of the probe's own dataset (PhysioNet, UBFC-Phys, Stress-Predict, Campanella) under the same model; *for the EPM-E4 clips, whose stress windows come from all five datasets, the pooled pair with the same stress windows. Δ ref: joint subject bootstrap within dataset; p_{holm} for Δ ref, Holm within each model. Deviations from 0.5 detectable with 80% power: 0.06–0.07 for fear and the trained exercise model, 0.12–0.22 for the other pairs (matched_arousal_per_dataset/, reviewer_reanalyses/).

(per_dataset_reference.csv). At matched arousal the external model still separates stress from baseline and rest strongly on WESAD (0.95 [0.89, 0.98]) but only modestly on PhysioNet (0.70), UBFC-Phys (0.68), Campanella (0.61) and Stress-Predict (0.57). A pooled five-dataset reference would read 0.58, 0.09 below the window-weighted mean of the per-dataset values, because external scores are not calibrated across datasets inside a bin; every comparison below therefore uses the probe's own dataset. Against that

reference no probed state is more separable than rest for the external model: exercise is +0.06 (CI −0.02 to +0.14), hyperventilation −0.05 and the UBFC-Phys control version −0.11, and Lego tasks and anger clips fall significantly below it. Once the probed state is a training negative, the UBFC-Phys control version stays significantly below its reference (−0.17), hyperventilation and Lego stay below it with CIs that include zero, and only exercise (+0.19) and the fear clips (+0.05) rise above it: outside WESAD, the model separates stress from rest beyond arousal only modestly, and not better from these demanding tasks. These are absence-of-evidence results. Only hyperventilation is equivalent to chance within ± 0.10 , and the tests could detect only deviations of 0.12–0.22 from 0.5 at 80% power. Under session scaling two states are separable. Exercise reaches 0.892 within bins once the model has seen it (against 0.5, $p_{\text{holm}} = 0.003$) and 0.759 even without, though then not above PhysioNet's own reference, and fear clips reach 0.779; the bootstrap difference between the exercise AUROC and each non-separable pair excludes zero (lower bounds 0.18–0.36). For that model, repeating the test with a heart-rate-only index gives 0.91 and with an EDA-only index 0.82, so exercise stays separable whichever channel defines arousal; it is separated by its pattern across channels, not by arousal level. (For the external model, which has not seen exercise, the same indices give 0.65 and 0.54.)

Robustness. We reran the test under raw and baseline-referenced scaling, on windows with usable pulse only, and with a synthetic positive control (matched_arousal_robustness/). The null pairs are robust: against the pooled reference used in these reruns, the UBFC-Phys control version, hyperventilation and Lego never exceed it under any scaling or on complete-case

windows. The separable pairs are not. Exercise's excess over the reference appears only under session scaling (external Δ ref -0.37 raw, -0.22 baseline-referenced), and the fear-clip excess vanishes on windows with usable pulse (Δ ref 0.00) but grows under raw scaling ($+0.18$). The positive control measures sensitivity: a stress-only shift of 0.1 SD in the model score is detected on the reference pair (3839 baseline and rest windows), but a 1-SD shift in a real feature (*eda_slope*) injected into half the stress subjects is not detected on the UBFC-Phys pair (80 control windows) ($\Delta = 0.04$, CI crossing zero). Only the large pairs can therefore support a null. Screening windows with the EDA rules of Kleckner et al. [29] (out-of-range level, too-fast change, and their surroundings; 0–11% of windows rejected per class) or with those rules plus their temperature rule (up to 53% of early-session baseline windows rejected, because the 30-°C floor tracks time in session) leaves every null pair within 0.03 of the pooled reference or below it and, under the EDA rules, pushes Lego further below (external within-bin AUROC 0.32), so motion artefacts in the Lego tasks do not explain it; external exercise and the fear clips, once trained, keep positive margins that are no longer significant under the full screen (*eda_quality_hr_arm/*).

Because the index is built from two model features, the test was repeated with index and model on disjoint channels (*partA_disjoint.csv*): a heart-rate-only index with an EDA-only model, and an EDA-only index with a cardiac-only model. Hyperventilation and the UBFC-Phys control version stay inseparable in every cell (0.45–0.54). The other pairs depend on the variant. Lego becomes separable under the heart-rate index (0.73 external, 0.84 once trained), where the pair shrinks to the 259 stress and Lego windows with usable pulse (63% of Campanella windows have none); anger becomes separable under the EDA index once trained (0.74); and with the EDA index the external cardiac-only model scores exercise above stress (AUROC 0.29), since within EDA bins exercise windows carry far higher heart rate (median heart-rate index 3.9 against 1.2). Removing only the three features closest to the index (*hr_mean*, *eda_tonic_mean*, *eda_mean*) leaves correlated EDA features in the model and changes no row by more than 0.09. Heart rate is also missing unevenly: in WESAD 63% of stress windows have no usable pulse against 10% of baseline windows, so the index falls back to EDA for most WESAD stress windows. Exercise windows sit almost entirely above every stress window (median index 4.8 against 0.8), so its matched bins cover only the lowest three quintiles.

E. Time in session and early-recording drift

Baseline is recorded first in every dataset, so any baseline-referenced comparison is confounded with time in session. Fig. 3(a) quantifies this: minutes since recording start, used as the only feature, reaches BA 0.83 on PhysioNet (HR/HRV/EDA model 0.78; better in 16 of 20 splits) and 0.95 on Stress-Predict (HR/HRV/EDA model 0.70; 20 of 20), ties on WESAD (0.90 against 0.92) and is perfect on Campanella. Given minutes as an extra feature, the within-dataset model exploits it (Stress-Predict 0.70 to 0.95) and

then collapses externally (0.66 to 0.56; WESAD 0.90 to 0.45), because protocol timing does not transfer. Any within-dataset pipeline that leaks time, through raw temperature, drift features or the absence of per-subject normalisation, inherits this. The HR/HRV/EDA model survives position-matched negatives with changes of at most -0.03 within and -0.04 externally (*time_probe_summary.csv*), so the LODO result is not riding on time; the matched negatives are mostly post-stressor rest, so this also bounds recovery contamination. Matching does not remove the time signal everywhere, however: under the same rule the time-only classifier falls to 0.60–0.62 on PhysioNet and WESAD but stays at 0.88 on Stress-Predict (HR/HRV/EDA model 0.67), and matching is impossible on Campanella, whose non-stress windows all precede its stressor, and on UBFC-Phys, whose per-task files carry no session clock (*rev9_position_matched_summary.csv*). The rule is necessary, not sufficient. A baseline-referenced feature set, by contrast, calls 50–83% of later non-stress windows stress across the WESAD–PhysioNet pair (*order_confound_check.csv*).

Part of the time signal is drift early in the recording. Measured from the start of each E4 recording (*warmup_since_dorning.csv*), subject-centred skin temperature rises at 0.02–0.08 °C/min in the first 10 min on all four stress datasets with a continuous clock. It keeps rising past 20 min on WESAD and rises until about minute 20 on Campanella before levelling off ($+1.8$ and $+0.9$ °C between minutes 1–5 and 30–40; 13 of 15 WESAD subjects warm over the first 30 min, by up to $+6.5$ °C). Tonic EDA rises in the first 10 min on all four. Baseline falls inside that phase on Campanella (100% of baseline windows within 10 min of recording start), Stress-Predict (97%) and PhysioNet (68% within 10 min, 90% within 15), and stress arrives at a median of 7–28 min. WESAD is the exception: its synchronised protocol starts 16–26 min after the raw E4 recording (median 21 min, found by matching the wrist EDA stream), so its baseline begins 17–49 min after the device was switched on (median 33 min) and no baseline window falls within 15 min of recording start. WESAD is also the one dataset on which the time-only classifier does not beat the HR/HRV/EDA model. Whether recording start equals donning is not documented by any dataset; on WESAD, every subject's first minute of temperature reads 4–6 °C above the following minutes, which suggests the device was placed within that minute. A recording without a stressor to anticipate helps separate the two drifts (*rev15_settling_control.csv*), although it is not stimulus-free. In EPM-E4, where participants watch emotional film clips from about minute 2, skin temperature does not rise over the first 20 min (-0.02 °C between minutes 1–4 and 15–19, against $+0.39$ to $+0.87$ °C on the four stress datasets), while tonic EDA drifts upward as on the stress datasets ($+0.69$ against $+0.33$ to $+0.59$ μ S). The EDA drift is therefore consistent with electrode and skin settling, although the clips may contribute, whereas the temperature rise goes with the stress protocols rather than with the device alone. This is the quantitative form of the sequence effect described by Richer et al. [9] and the reason temperature features track

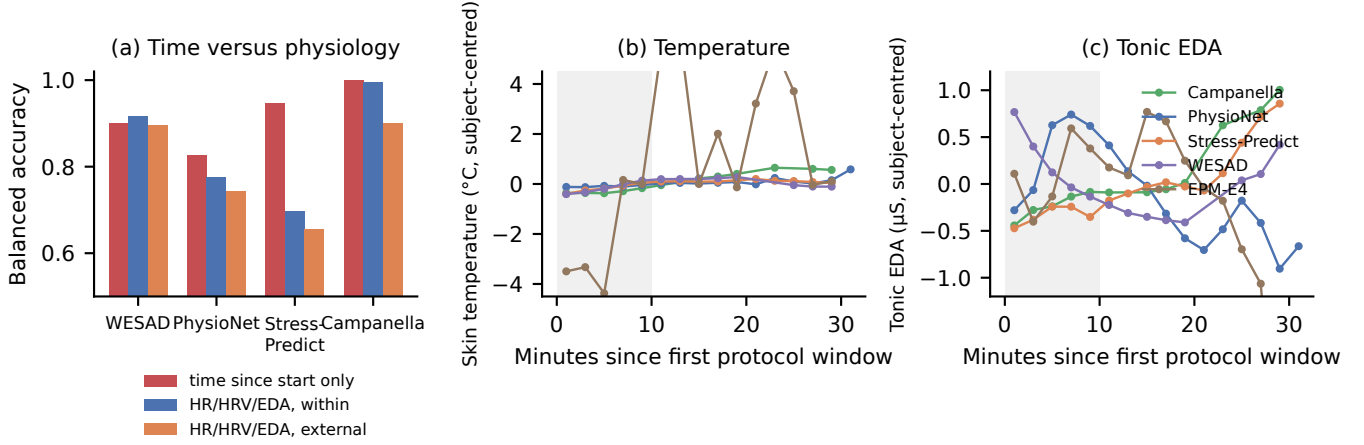


Fig. 3. (a) Balanced accuracy of a classifier that sees only minutes since recording start, against the HR/HRV/EDA model within and across datasets. (b), (c) Subject-centred skin temperature and tonic EDA against minutes since the first protocol window (2-min bins); the shaded band marks the first 10 min, in which 100% of Campanella, 97% of Stress-Predict and 68% of PhysioNet baseline windows fall when measured from recording start (90% within 15 min). WESAD’s protocol clock starts 16–26 min after its recording; EPM-E4’s starts at recording.

time and are excluded from the main feature set.

F. Error budget and negative results

Four test-side filters for recovery contamination, sensor loss, onset latency and non-response (the last uses test labels and is diagnostic only) buy 0.05–0.06 BA on PhysioNet and Stress-Predict (0.02 for the external model on Stress-Predict). The three label-free filters buy at most 0.056 (upper CI bound 0.09), and the residual error of 0.17–0.23 (0.33 external on Stress-Predict) is explained by none of them. On WESAD, replacing wrist cardiac features with chest-ECG features on the same windows raises BA from 0.916 to 0.956, so motion-robust wrist heart rate is worth at most about +0.04 there. Unsupervised domain adaptation, few-shot personalisation with first-in-time calibration windows and per-user thresholds give no significant gain, self-rated stress intensity does not predict which windows are missed, and detectability is not a stable subject trait. Supplementary Sections S-I and S-II give the budget, the responder analysis and the full table of negative results.

VI. DISCUSSION

Wrist stress detectors behave as arousal detectors. Three results point the same way. An untrained index of heart rate and tonic EDA comes within 0.075 BA of the trained model on every unseen dataset. At matched arousal, a model that has not seen the target dataset separates stress from baseline and rest only modestly on four of five datasets (0.57–0.70), and no better from the lower-demand social task, hyperventilation, Lego assembly or anger clips. And the one clean non-stress state with high arousal, exercise, is called stress until it is shown as a negative. The distribution of that arousal is similar across laboratory datasets, so transfer is cheap and domain adaptation has little to align. This is a statement about what stage-labelled training extracts from wrist HR, HRV and EDA in these paradigms, not a claim that stress has no autonomic signature.

It is consistent, by extrapolation from laboratory emotion research, with a meta-analysis of 202 emotion-induction studies in which the pattern of autonomic effect sizes “did not clearly distinguish one emotion category from another” [31]. Two readings remain open and the present data cannot separate them: the detector may have learned only arousal, or the wrist channels may carry nothing more: psychosocial stress activates the hypothalamic–pituitary–adrenal axis as well as the sympathetic system, and that axis is assessed through cortisol [9], which no wrist channel measures; across 208 laboratory studies, tasks “containing both uncontrollable and social-evaluative elements were associated with the largest cortisol and adrenocorticotropin hormone changes” [32], while wrist EDA tracks palmar EDA only moderately [21]. WESAD is the exception, and it agrees with Kwon et al. [5]: with their within-subject heart-rate matching our model also keeps WESAD stress separable from baseline (AUROC 0.98), and even joint heart-rate and EDA matching leaves it at 0.93–0.95 (kwon_reconciliation/). The TSST, the most potent stressor here, carries information beyond arousal magnitude on the wrist; the milder stressors of the other four datasets carry little. Exercise separates from stress at matched arousal only under session scaling, and the fear-clip margin vanishes on windows with usable pulse, so neither separability is a robust property of the wrist signal; what is robust for exercise is that its false-stress rate falls under every scaling once it is a training negative. The datasets with low ceilings are those whose stressors produce a sixth (PhysioNet) to a third (Stress-Predict) of the electrodermal response of the TSST in WESAD; filters for recovery, sensor loss and onset recover at most 0.06 of that gap, and a strict responder filter at most 0.09 on PhysioNet and nothing on Stress-Predict. We read the ceiling as set by the size of the physiological response a protocol provokes. Because that size is measured with the same channels, this is a description, not an independent test of stressor potency. The few-shot and domain-adaptation nulls are consistent with it: personalisation cannot recover a response

TABLE V
PROPOSED REPORTING CHECKLIST FOR WEARABLE STRESS DETECTORS

Item	What to report
Splits	Subject-grouped splits; leave-one-dataset-out when several datasets are pooled; corrected paired tests (e.g. Nadeau–Bengio) with CIs on every difference.
Index-only baseline	BA and AUROC of an untrained HR+EDA arousal index under the same splits, and the model-minus-index difference with its CI.
Specificity panel	False-stress rate with subject-bootstrap CI on exercise, on a lower-demand version of each stressor task, and on strong emotion; subjects and windows per state; rate at the reported threshold and at fixed sensitivity.
Matched-arousal test	Index defined before the analysis, ideally on channels the model does not use; the same-dataset stress-versus-rest reference pair; Δ ref with CI; an equivalence margin (e.g. ± 0.10) and the minimum detectable deviation.
Protocol position	Time-only classifier BA; results under position-matched negatives whenever baseline precedes the stressor; time since donning or a settling period before baseline.
Missing data	Fraction of windows without usable pulse, by class.

that did not occur.

Deployment. A detector that responds to arousal will raise alerts during any arousing activity. As an illustration, at a stress prevalence of 5% and recall 0.75, false-stress rates of 0.10, 0.33 and 0.69, spanning Table III, give positive predictive values of 0.28, 0.11 and 0.05. Outputs of such detectors are better labelled “arousal” than “stress” until a specificity panel says otherwise. Consumer devices without EDA fare worse: a heart-rate and HRV-only model transfers at chance to Campanella (BA 0.50) and weakly to Stress-Predict (0.56), calls 96–97% of exercise windows stress when exercise is not in training, and still 45% when it is.

Reporting norms. We propose that a wearable stress detector be published with the checks in Table V, three of them new: an index-only baseline, reporting how far the trained model beats an untrained HR+EDA index under the same split; a specificity panel, its false-stress rate on exercise, on lower-demand versions of its stressor tasks and on strong emotion, with CIs and at least about 20 subjects per state, the way an assay reports cross-reactivity; and a matched-arousal test against a reference pair, with a stated equivalence margin, where such states are available. We also propose three protocol rules: a settling period before baseline, or time since donning as a covariate; position-matched negatives, or at least a report of the time-only classifier, whenever baseline precedes the stressor; and exercise among the training negatives, or an activity gate, whenever the deployment includes movement.

What a decisive study needs. No public E4 dataset separates speech from evaluation within subject or counterbalances order; UBFC-Phys contrasts two demand levels of a social task between subjects. The friendly TSST, built to match the TSST in structure and cognitive demand with the social-evaluative elements removed, and run within subject in randomised order by Richer et al. [9], is the control such a study needs on the wrist. A within-subject design with speech and evaluation crossed, cycling at matched heart rate, cognitive load without feedback, Latin-square order with baselines at both ends, chest ECG and a palmar EDA reference would settle whether wrist EDA carries an evaluative-threat signature that the present data cannot resolve, and would provide the first order-matched

baseline on the wrist.

Limitations. Whole-session per-subject normalisation is transductive; raw and causal variants keep the small transfer cost on long recordings but lose about 0.10 on short ones. The probed states are not clean non-stress controls apart from exercise and the passive clips, the hyperventilation, anger and UBFC-Phys control cells are small, and the UBFC-Phys contrast is between subjects; the matched-arousal nulls exclude only deviations above 0.12–0.22. Mean skin-conductance level from the E4 correlated with palmar EDA at $r = 0.61$ and 0.40 in two resting baselines and not significantly ($r = 0.30$) during conversation [21], so a wrist null on a stress-specific signature may be instrument failure. The arousal index is built from two model features; the channel-disjoint variants keep hyperventilation and the UBFC-Phys control inseparable but make Lego or anger separable in one variant each, and heart rate is missing far more often in some stress classes (63% of WESAD stress windows) than in their baselines. Responder filters use test labels and are diagnostic only; the strict definitions discard half the subjects. All data are laboratory recordings; ambulatory validity is untested. The pooled subject count (132 in the five stress datasets) is small for a field-level claim, and the upper CI bounds of the transfer costs reach 0.08–0.13.

VII. CONCLUSION

Across five public laboratory E4 datasets with audited labels, a wrist stress detector built on HR, HRV and EDA features transfers to an unseen dataset at a cost of 0.015–0.05 BA, with no cost significant and upper bounds of 0.08–0.13, and none of the methods tested here, from domain adaptation to few-shot calibration, improved on that. An untrained index of heart rate and tonic EDA comes within 0.075 BA. At matched arousal the detector separates stress from rest strongly only on WESAD (0.95), modestly elsewhere (0.57–0.70), and no better from lower-demand social tasks, hyperventilation, manual work or anger clips, and it calls exercise stress until exercise is in training. On these data, wrist stress detectors behave as arousal detectors. Protocol position predicts stress better than physiology on the hardest datasets, whose baselines are recorded in the first minutes, while the signals still drift, and the within-dataset ceiling tracks the size of the physiological response each protocol provokes rather than the pipeline. The next gains are more likely to come from protocol design and contextual sensing than from classifiers. Two studies follow from this work: a counterbalanced within-subject protocol that crosses speech with evaluation, adds matched-heart-rate exercise and non-evaluative cognitive load, and records chest ECG and palmar EDA alongside the wrist, to test whether wrist EDA carries an evaluative-threat signature at all; and an ambulatory validation in which the specificity panel is scored against labelled daily-life arousal, which no laboratory dataset can provide.

ETHICS STATEMENT

This work analyses only public, de-identified datasets, each recorded under the ethics approval of its original study; no new data were collected and no participant was contacted.

DATA AND CODE AVAILABILITY

The five datasets are available from their original sources under their own licences. The audited label tables, the 20 frozen subject-grouped splits, the feature-extraction and evaluation scripts, and the scripts for the specificity panel, matched-arousal test, time-in-session probe and error budget are released at <https://github.com/ALVAROI12/smartwatch-stress-detection> (branch `jbhi-revision`).

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